Project 4:-Maven Selenium Project – Automating Web Applications Using Firefox Browser

Objective

To create a **Maven-based Selenium project using the Firefox browser** that performs the following tasks:

- Login to the SauceDemo website
- Search and add a product to the cart on AutomationExercise website
- Login to the Practice Test Automation website
- Demonstrate **multi-tab browser automation using Selenium WebDriver**

Websites Used for Automation

The following websites are used in this project:

1. SauceDemo
   https://www.saucedemo.com/
2. Automation Exercise
   https://automationexercise.com/products
3. Practice Test Automation
   https://practicetestautomation.com/practice-test-login/

Step 1: Create a Maven Project

Open the terminal and create a Maven project using the following command:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumFirefox -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
brijesh@Ubuntu:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumFirefox -DarchetypeA
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/brijesh/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumFirefox
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/brijesh/devops/MyMavenSeleniumFirefox
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.316 s
[INFO] Finished at: 2026-03-21T13:36:07+05:30
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/devops$
```

```
brijesh@Ubuntu:~/devops$ cd MyMavenSeleniumFirefox
brijesh@Ubuntu:~/devops/MyMavenSeleniumFirefox$ tree
.
├── pom.xml
└── src
    ├── main
    │   └── java
    │       └── com
    │           └── example
    │               └── App.java
    └── test
        └── java
            └── com
                └── example
                    └── AppTest.java

9 directories, 3 files
brijesh@Ubuntu:~/devops/MyMavenSeleniumFirefox$
```

Step 2: Install Firefox Browser

Check whether Firefox is installed.

firefox --version

Example output:

Mozilla Firefox 124.0

If Firefox is not installed:

sudo apt install firefox

Step 3: Install GeckoDriver

Selenium requires **GeckoDriver** to control Firefox.

Download GeckoDriver:

wget https://github.com/mozilla/geckodriver/releases/latest/download/geckodriver-v0.34.0-linux64.tar.gz

Extract the file:

tar -xvzf geckodriver-v0.34.0-linux64.tar.gz

Move it to system path:

sudo mv geckodriver /usr/local/bin/
sudo chmod +x /usr/local/bin/geckodriver

Verify installation:

geckodriver --version

Step 4: Add Selenium Dependency in pom.xml

Open the **pom.xml** file and add the Selenium dependency.

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>
<artifactId>MyMavenFirefoxApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>
```

```
<name>MyMavenFirefoxApp</name>
<dependencies>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
<configuration>
<source>11</source>
<target>11</target>
</configuration>
</plugin>
<plugin>
<groupId>org.codehaus.mojo</groupId>
<artifactId>exec-maven-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<mainClass>com.example.App</mainClass>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 5:

Inspect Elements in the Websites

Before writing Selenium code, inspect the webpage elements.
Steps:

1. Open the website.
2. Right-click on the webpage.
3. Click **Inspect**.
4. Use the **cursor tool** to identify HTML elements.

Step 6: Create App.java and add the following code

Open | *App.java ~/Devops/MyMavenSeleniumFirefox/src/main/java/com/example | Save

```java
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;

import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;

import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.JavascriptExecutor;

import org.openqa.selenium.support.ui.WebDriverWait;
import org.openqa.selenium.support.ui.ExpectedConditions;

import org.openqa.selenium.WindowType;

import java.time.Duration;

public class App {

    public static void main(String[] args) {


        FirefoxOptions options = new FirefoxOptions();
        options.setBinary("/usr/bin/firefox");
        options.addArguments("--no-sandbox");

        WebDriver driver = new FirefoxDriver(options);
        driver.manage().window().maximize();

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));

        try {

            // -------- Tab 1 : SauceDemo --------
            driver.get("https://www.saucedemo.com/");

            wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("user-name")))
                    .sendKeys("standard_user");

            driver.findElement(By.id("password")).sendKeys("secret_sauce");
            driver.findElement(By.id("login-button")).click();

            // -------- Tab 2 --------
            driver.switchTo().newWindow(WindowType.TAB);
```

Java | Tab Width: 8 | Ln 26, Col 48 | INS

```
Open ▾ *App.java ~/Devops/MyMavenSeleniumFirefox/src/main/java/com/example Save
52                  ExpectedConditions.visibilityOfElementLocated(
53                          By.xpath("(//div[@class='product-image-wrapper'])[1]")
54                  )
55          );
56
57          actions.moveToElement(product).perform();
58
59          WebElement addToCart = wait.until(
60                  ExpectedConditions.elementToBeClickable(
61                          By.xpath("(//a[contains(text(),'Add to cart')])[1]")
62                  )
63          );
64
65          JavascriptExecutor js = (JavascriptExecutor) driver;
66          js.executeScript("arguments[0].scrollIntoView(true);", addToCart);
67
68          addToCart.click();
69
70          WebElement continueBtn = wait.until(
71                  ExpectedConditions.elementToBeClickable(
72                          By.xpath("//button[text()='Continue Shopping']")
73                  )
74          );
75
76          continueBtn.click();
77
78          // -------- Tab 3 --------
79          driver.switchTo().newWindow(WindowType.TAB);
80          driver.get("https://practicetestautomation.com/practice-test-login/");
81
82          wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("username")))
83                  .sendKeys("student");
84
85          driver.findElement(By.id("password")).sendKeys("Password123");
86          driver.findElement(By.id("submit")).click();
87
88          // Optional validation
89          wait.until(ExpectedConditions.urlContains("logged-in-successfully"));
90
91      } catch (Exception e) {
92          e.printStackTrace();
93      } finally {
94          driver.quit();
95      }
96  }
97 }
Java ▾ Tab Width: 8 ▾ Ln 26, Col 48 ▾ INS
```

Step 7:

Compile and build the project using the command:

mvn clean install

```
brijesh@Ubuntu:~/Devops/MyMavenSeleniumFirefox$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------< com.example:MyMavenSeleniumFirefox >-----------------
[INFO] Building MyMavenSeleniumFirefox 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumFirefox ---
[INFO] Deleting /home/brijesh/Devops/MyMavenSeleniumFirefox/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumFirefox ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenSeleniumFirefox/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumFirefox ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenSeleniumFirefox/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumFirefox ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenSeleniumFirefox/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumFirefox ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenSeleniumFirefox/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumFirefox ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.012 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.0.1:jar (default-jar) @ MyMavenSeleniumFirefox ---
[INFO] Building jar: /home/brijesh/Devops/MyMavenSeleniumFirefox/target/MyMavenSeleniumFirefox-1.0-SNAPSHOT.jar
[INFO]
```

Step 8: Run the Application

Run the application using:

```
brijesh@Ubuntu:~/Devops/MyMavenSeleniumFirefox$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------< com.example:MyMavenSeleniumFirefox >-----------------
[INFO] Building MyMavenSeleniumFirefox 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenSeleniumFirefox ---
```